

Computer Network Complex Computational Problem

Objective:

The objective of this lab is to design and implement a basic network infrastructure for a small office, integrating both wired and wireless components. Analyze the network's performance in handling everyday tasks and optimize for efficiency and reliability.

Theory:

What is a network?

- A **network** is a group of interconnected devices (such as computers, printers, smartphones) that communicate with each other to share data, resources (like printers), and services (such as the internet or file servers). Networks can be small (like a home or office) or large (like the internet).

What are wired and wireless networks?

- **Wired Network:** Uses physical cables (e.g., Ethernet) to connect devices. It offers higher speed, security, and reliability but limits mobility.
- **Wireless Network:** Uses radio signals (e.g., Wi-Fi) to connect devices without cables. It offers flexibility and mobility but may suffer from signal interference and lower speeds compared to wired networks.

Why routers switches and access points are used?

- **Router:** Connects different networks together, such as your local office network to the internet. It also manages IP addresses and routes data between devices.
- **Switch:** Connects multiple wired devices within the same network (e.g., PCs, printers) and forwards data intelligently based on MAC addresses.
- **Wireless Access Point (AP):** Extends the network by providing wireless connectivity to Wi-Fi-enabled devices. It acts as a bridge between the wired network and wireless clients.

What is DHCP? IP Addressing, and Network Topologies

- **DHCP** is a network service that automatically assigns IP addresses to devices on the network, saving time and reducing configuration errors.

What is IP Addressing?

An **IP address** is a unique identifier assigned to each device on a network. It allows devices to communicate with each other. IPs can be assigned **dynamically** (by DHCP) or **statically** (manually).

What are Network Topologies?

This refers to the physical or logical layout of a network. Common topologies include:

- **Star Topology** (most common): All devices connect to a central switch or router.
- **Bus Topology**: All devices share a single communication line (rare today).
- **Mesh Topology**: Every device connects to every other (used in high-reliability systems).
- **Ring Topology**: All devices are connected in loop creating a ring.

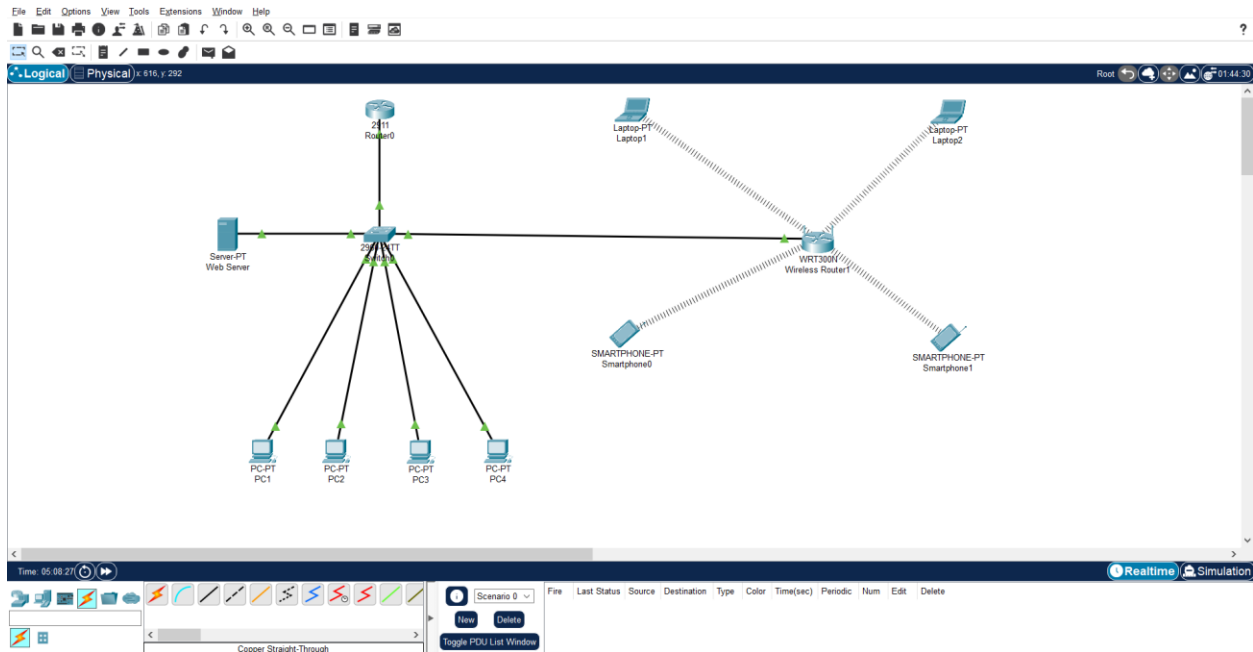
Network Description:

We are using a **Star Topology** for this small office network.

In this topology:

- All **wired devices** connect to a central **switch**.
- The **switch** connects to a **router**, which provides internet access and DHCP services.
- A **wireless access point (AP)** connects to the switch and serves **wireless devices** like laptops and phones.
- All devices (wired and wireless) receive IP addresses from the **router's DHCP server**.

Network Design:



Devices used in Small Office Network:

Device	Quantity	Description
Router	1	Provides IP addressing and internet access
Switch	1	Connects wired devices
Wireless Access Point	1	Enables Wi-Fi for laptops, phones
Server (Web Server)	1	Used to access the web service
PCs (Wired)	4	Office desktops
Laptops (Wireless)	2	Mobile users
Smartphones (Wireless)	2	Mobile devices

Device Connection Table with Interfaces:

From Device	From Interface	To Device	To Interface
Router (2911)	GigabitEthernet0/0	Switch (2960)	FastEthernet0/1
Switch (2960)	FastEthernet0/2	PC1	FastEthernet0
Switch (2960)	FastEthernet0/3	PC2	FastEthernet0
Switch (2960)	FastEthernet0/4	PC3	FastEthernet0
Switch (2960)	FastEthernet0/5	PC4	FastEthernet0
Switch (2960)	FastEthernet0/6	Wireless Router	Ethernet 1
Switch (2960)	FastEthernet0/7	Server (Web Server)	FastEthernet0
Wireless Router	Wireless Interface	Laptop1	Wireless0
Wireless Router	Wireless Interface	Laptop2	Wireless0
Wireless Router	Wireless Interface	Phone1	Wireless0
Wireless Router	Wireless Interface	Phone2	Wireless0

IP Addressing Scheme

We'll use the private IP range **192.168.1.0/24**, which supports up to 254 hosts.

Device/Function	IP Address	Method	Notes
Router (Gateway)	192.168.1.1	Static	Default gateway for the LAN
Access Point	192.168.1.2	Static	Manually assigned
Server (Web Server)	192.168.1.5	Static	Manually assigned
PCs (Wired)	192.168.1.10-13	DHCP	Assigned automatically
Smartphones	192.168.1.14-15	DHCP	Assigned automatically
Laptops (Wireless)	192.168.1.16-17	DHCP	Assigned automatically
DHCP Range	192.168.1.10-254	DHCP	Pool set in router config

Subnet Mask	255.255.255.0	—	Standard Class C subnet
Default Gateway	192.168.1.1	—	Router's LAN IP

Router Configuration:

To assign the **IP address** to the router interface:

Router>en

Router#config t

Router(config)#interface gigabitEthernet0/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shutdown

To enable the **DHCP** on router:

Router(config)#ip dhcp pool Office

Router(dhcp-config)#network 192.168.1.0 255.255.255.0

Router(dhcp-config)#default-router 192.168.1.1

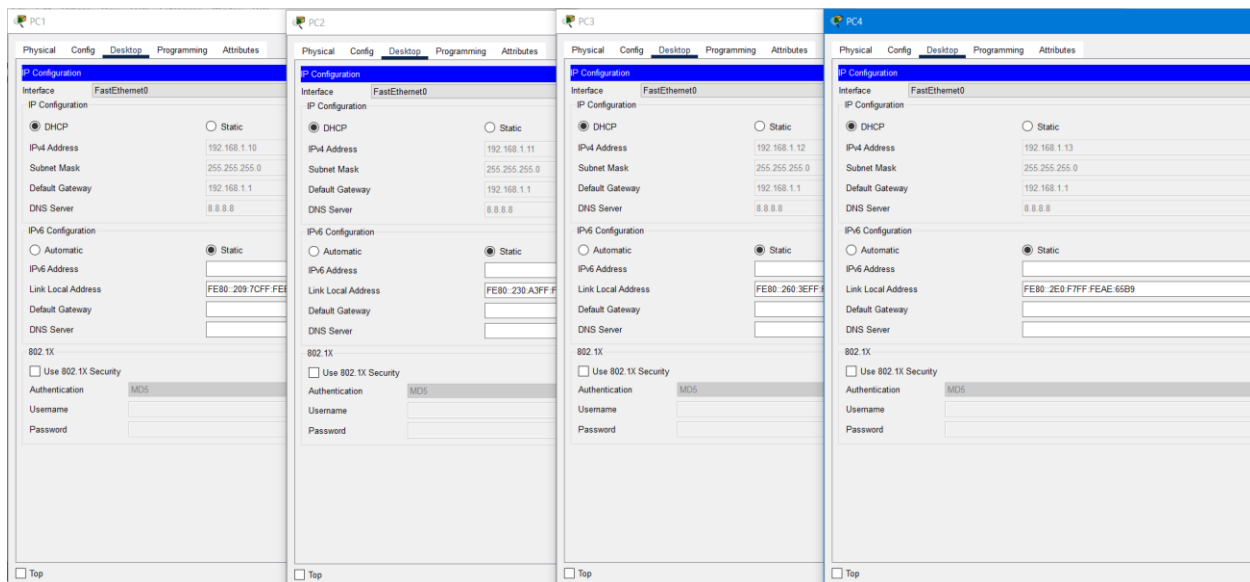
Router(dhcp-config)#dns-server 8.8.8.8

Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9

PCs Configuration:

1. Click on the **PC** (e.g., PC1)
2. Go to **Desktop > IP Configuration**
3. Select **DHCP**

The PC will now receive an IP address automatically from the **Router's DHCP server**.



PC Configuration		
IPv4 Address	Subnet Mask	Default Gateway
PC1		
192.168.1.10	255.255.255.0	192.168.1.1
PC2		
192.168.1.11	255.255.255.0	192.168.1.1
PC3		
192.168.1.12	255.255.255.0	192.168.1.1
PC4		
192.168.1.13	255.255.255.0	192.168.1.1

Wireless Router (WRT300N) Configuration:

1. Click on the **WRT300N Wireless Router**.
2. Go to the **GUI tab**.

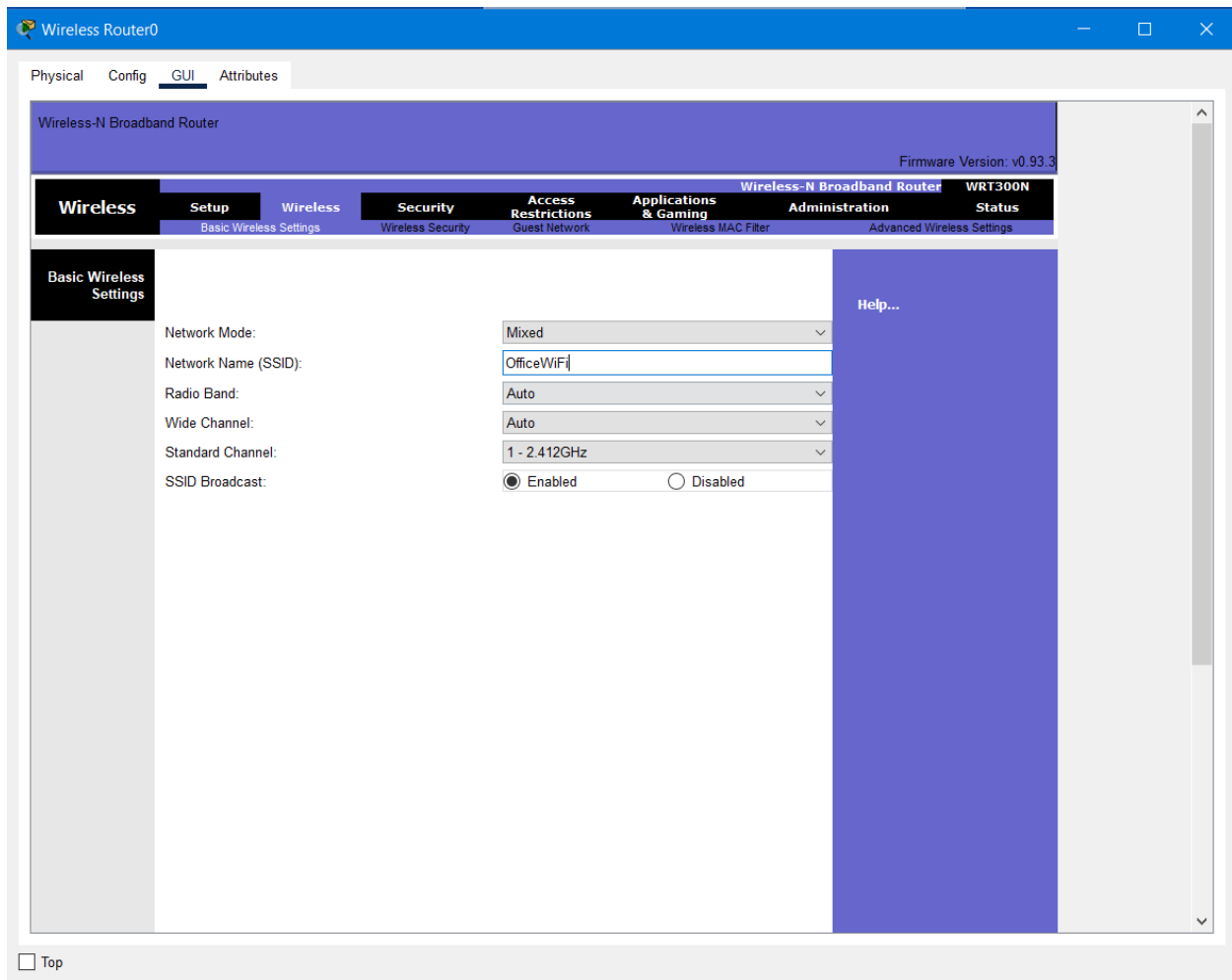
Basic Setup Tab:

Setting	Value
Internet Connection Type	Static IP
Internet IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1 (Router)
DHCP Server	Disable

The screenshot displays the 'Wireless Router0' web interface. The top navigation bar includes 'Physical', 'Config', 'GUI' (selected), and 'Attributes'. Below this, a sub-header reads 'Wireless-N Broadband Router' with 'Firmware Version: v0.93.3' on the right. A main menu bar contains 'Setup', 'Wireless', 'Security', 'Access Restrictions', 'Applications & Gaming', 'Administration', and 'Status'. The 'Setup' tab is active, showing a 'Basic Setup' sub-tab. The 'Internet Setup' section is expanded, showing 'Internet Connection type' set to 'Static IP'. Below this, fields for 'Internet IP Address' (192.168.1.2), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (192.168.1.1), and three 'DNS' fields (all set to 0) are visible. 'Optional Settings' include 'Host Name', 'Domain Name', and 'MTU' (Size: 1500). The 'Network Setup' section shows 'Router IP' (192.168.1.2) and 'Subnet Mask' (255.255.255.0). Under 'DHCP Server Settings', the 'DHCP Server' is set to 'Disabled' (radio button selected), and 'Start IP Address' is 192.168.1.100. A 'Top' link is at the bottom left.

Setting	Value
Internet Connection Type	Static IP
Internet IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS 1	0
DNS 2 (Optional)	0
DNS 3 (Optional)	0
Host Name	
Domain Name	
MTU	1500
Router IP	192.168.1.2
Subnet Mask	255.255.255.0
DHCP Server	Disabled
Start IP Address	192.168.1.100
Maximum number of Users	50

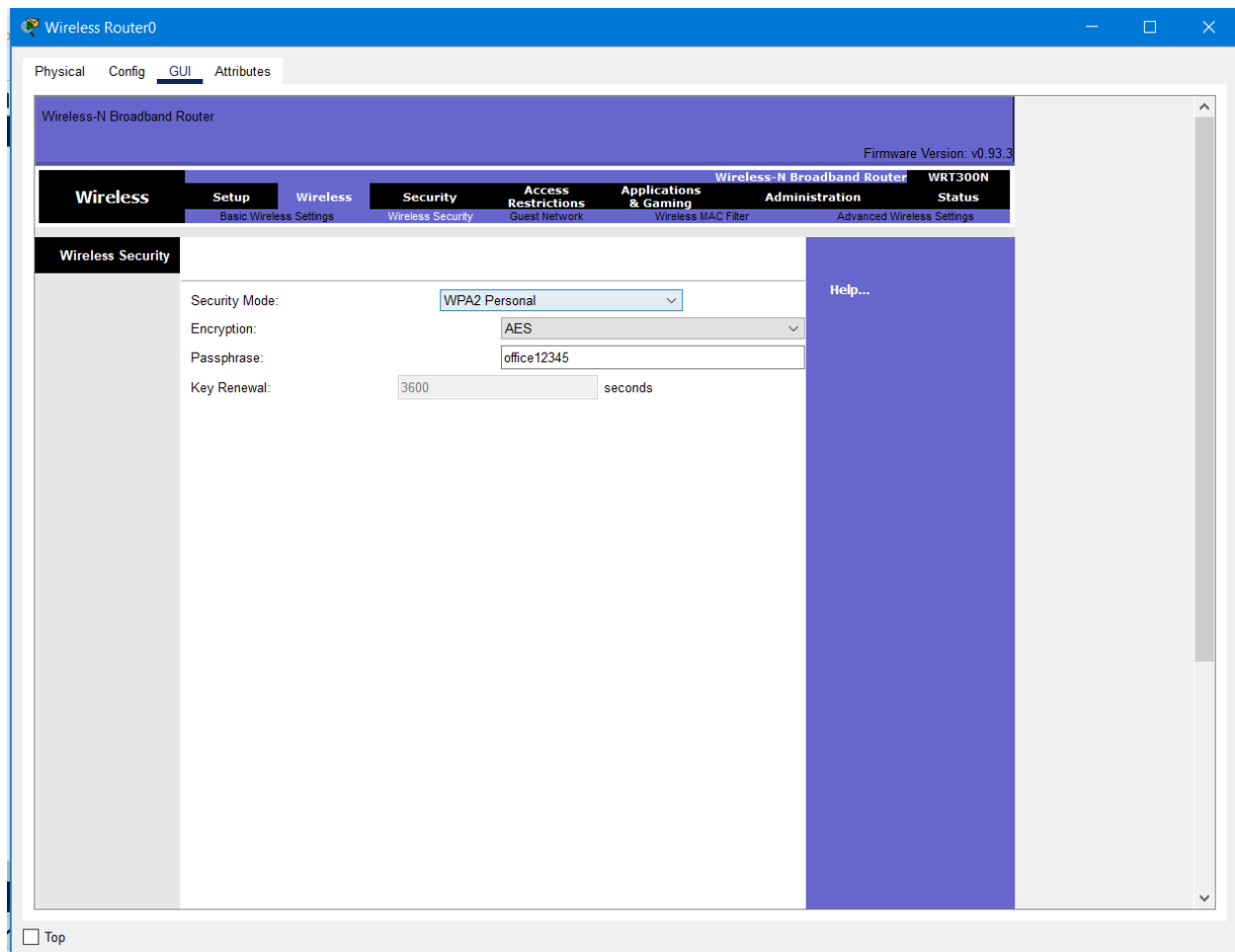
In Wireless Section:



1. **Network Mode:** Mixed
2. **Network Name (SSID):** OfficeWiFi
3. **Channel:** Auto

In Wireless Security Section:

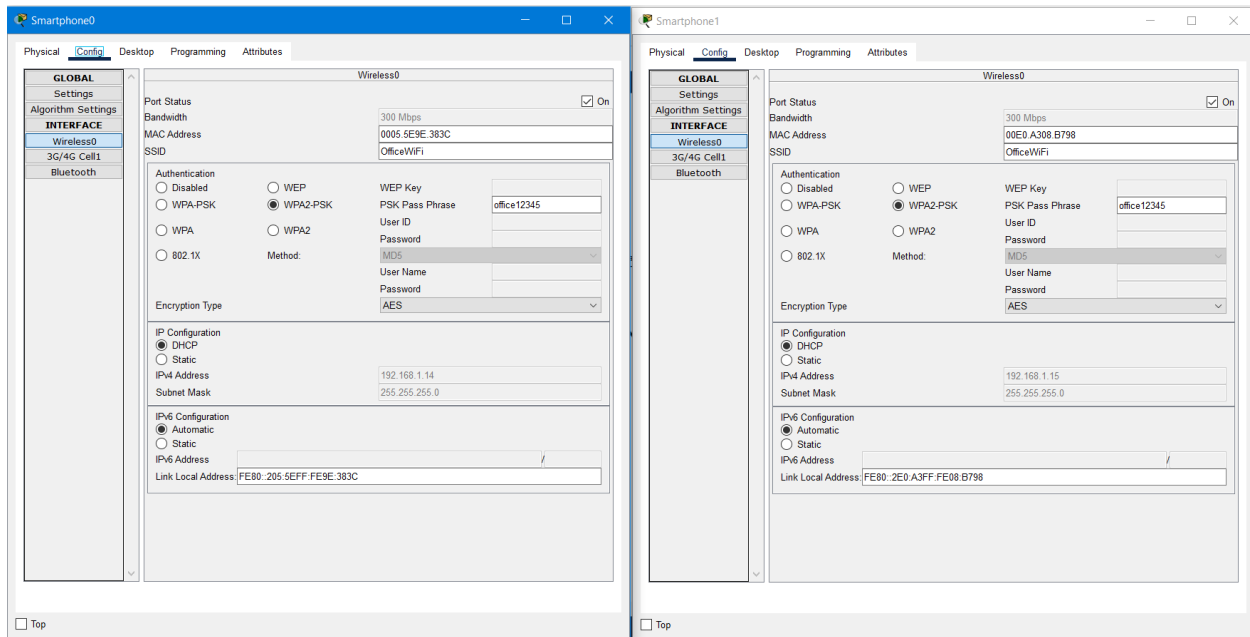
1. **Security Mode:** WPA2 Personal
2. **Encryption:** AES
3. **Passphrase:** office12345



Configuring Smartphones and Laptops:

Smartphone Setup (Phone1, Phone2):

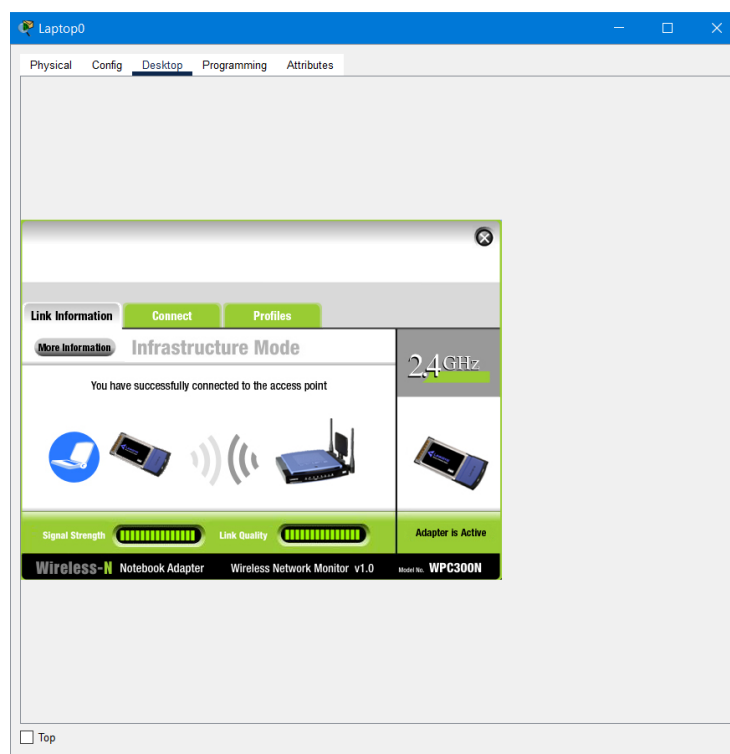
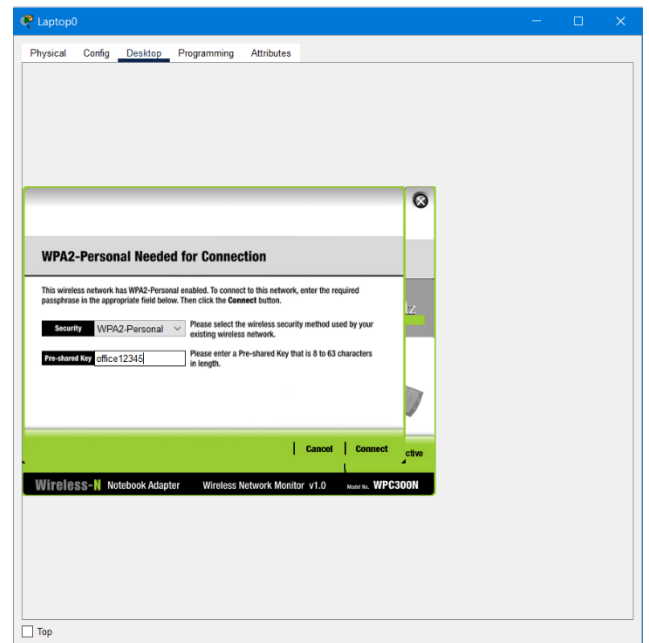
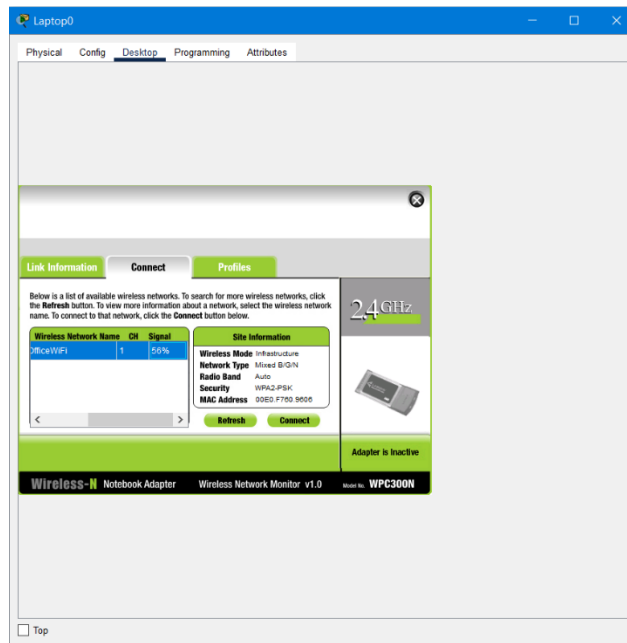
- Config tab → Wireless0
- Set SSID: OfficeWiFi
- Entered password: office12345
- Set IP Configuration to DHCP
- Successfully received IP addresses 192.168.1.14, 192.168.1.15



Laptop Setup (Laptop1, Laptop2):

Connect to the Wireless Network

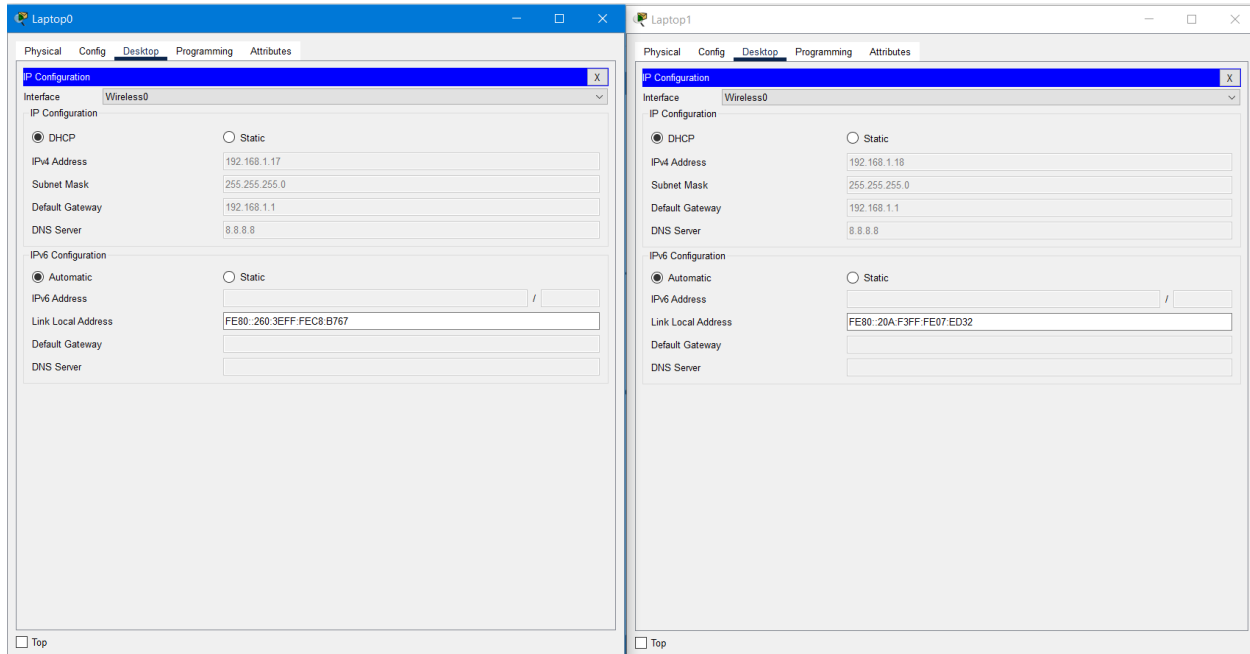
- Go to the Desktop tab → PC Wireless
- Click Connect
- Select your SSID (e.g., OfficeWiFi)
- Enter the WPA2 password (e.g., office12345)
- Wait for the status to show "Connected"



Same the process for another laptop.

Enable DHCP

- Still in Desktop tab, go to IP Configuration
- Set IP Configuration to DHCP
- The laptops should receive an IP address 192.168.1.16 and 192.168.1.17



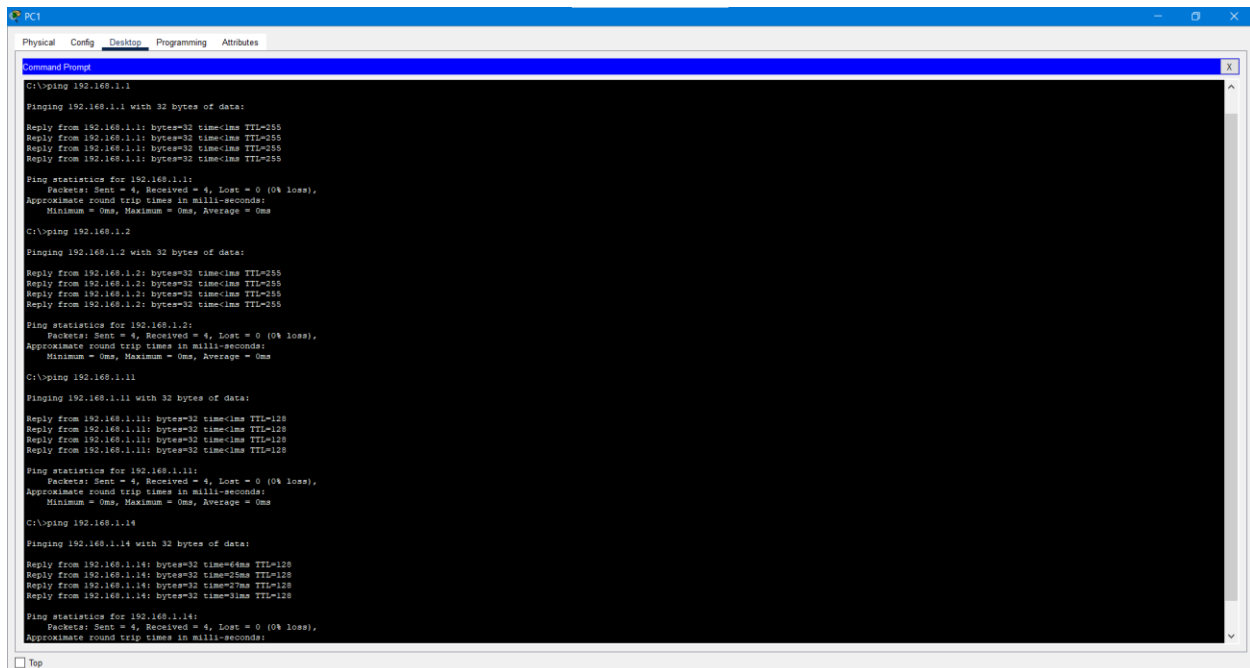
Smartphones and Laptops Configuration		
IPv4 Address	Subnet Mask	Default Gateway
Smartphone 1		
192.168.1.14	255.255.255.0	192.168.1.1
Smartphone 2		
192.168.1.15	255.255.255.0	192.168.1.1
Laptop 1		
192.168.1.17	255.255.255.0	192.168.1.1
Laptop 2		
192.168.1.18	255.255.255.0	192.168.1.1

Performance Analysis:

1. Connectivity:

From PC1 (192.168.1.10):

- ping 192.168.1.1 → Successfully reached the default gateway (Cisco router)
- ping 192.168.1.2 → Successfully reached the wireless access point (WRT300N)
- ping 192.168.1.11 → Communicated with PC2
- ping 192.168.1.16 → Reached Laptop1 (wireless)



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=1ms TTL=255
Reply from 192.168.1.2: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

From Laptop1 (192.168.1.16):

- ping 192.168.1.1 → Reached the main router
- ping 192.168.1.10 → Communicated with PC1 (wired)
- ping 192.168.1.17 → Verified communication with Laptop2 (wireless)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 5ms, Average = 5ms

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=6ms TTL=255
Reply from 192.168.1.1: bytes=32 time=3ms TTL=255
Reply from 192.168.1.1: bytes=32 time=3ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 4ms

C:\>ping 192.168.1.18

Pinging 192.168.1.18 with 32 bytes of data:

Reply from 192.168.1.18: bytes=32 time=9ms TTL=128
Reply from 192.168.1.18: bytes=32 time=7ms TTL=128
Reply from 192.168.1.18: bytes=32 time=3ms TTL=128
Reply from 192.168.1.18: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 9ms, Average = 5ms

C:\>
```

Results:

- Replies received = success
- Time $\approx$ 1–2 ms, 0% packet loss

2. Internet Speed:

To measure the internet speed on wired and wireless devices, a server will be added that will be used as a **Web Server**.

A static IP address was configured on the server:

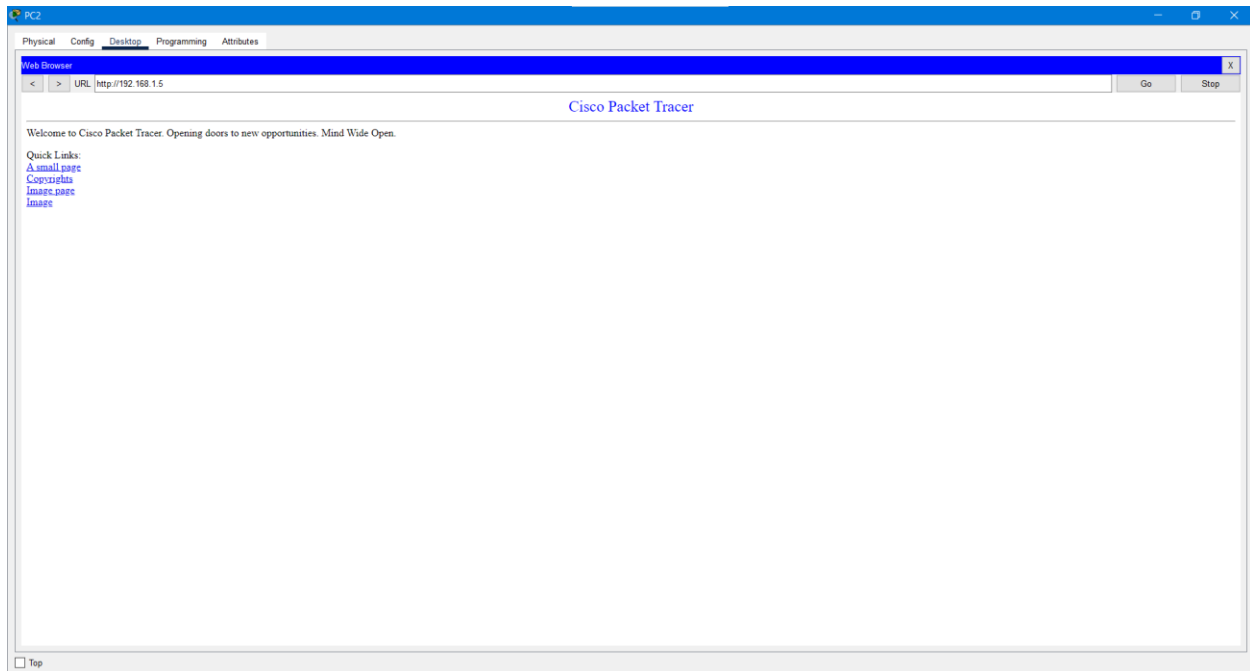
Setting	Value
IP Address	192.168.1.5
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

The HTTP service was enabled on the server:

- Server tab $\rightarrow$ Services $\rightarrow$ HTTP $\rightarrow$ Service: ON

On PC2:

- Opened Desktop tab → Web Browser
- Typed: `http://192.168.1.5`
- The server's web page loaded instantly



Result:

- Web browsing to the server's IP address confirmed that the HTTP service was operational and the network allowed smooth internal communication.
- No delay or packet loss was experienced, simulating a fast and reliable intranet service.
- This verifies proper client-server interaction within the LAN.

3. Wireless Coverage:

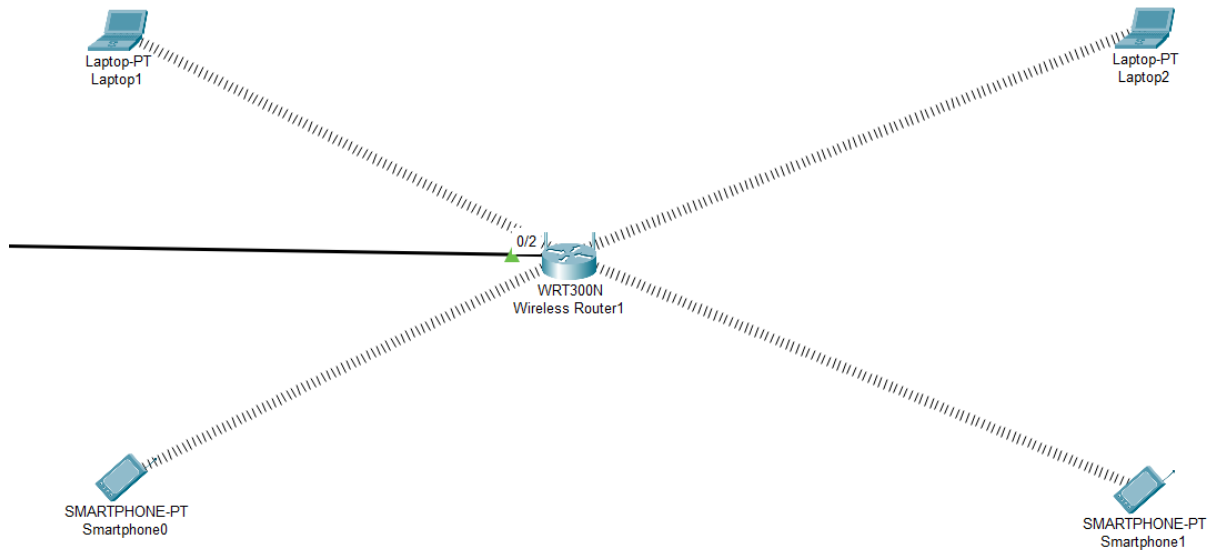
Evaluate how well wireless devices connect based on their distance from the Access Point.

How to Perform:

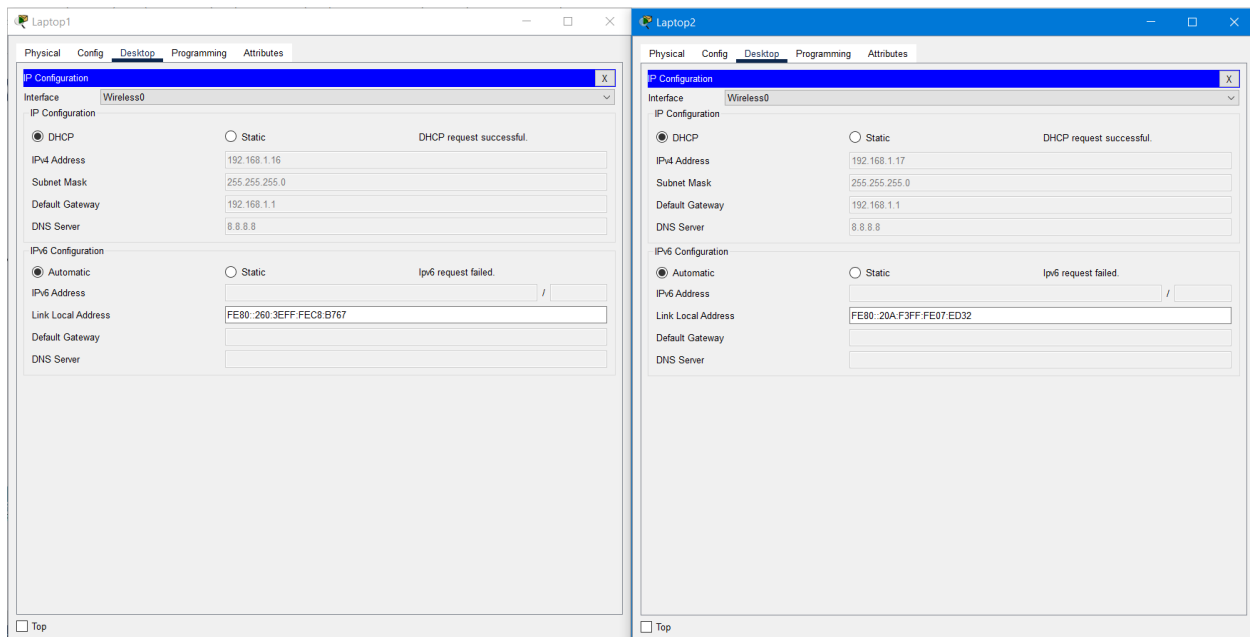
- Move laptops or smartphones around the workspace (in the topology)
- Reconnect to Wi-Fi (Config tab → Wireless0)

Observe:

1. If signal is still strong
2. If IP address is assigned via DHCP
3. If ping to 192.168.1.1 works

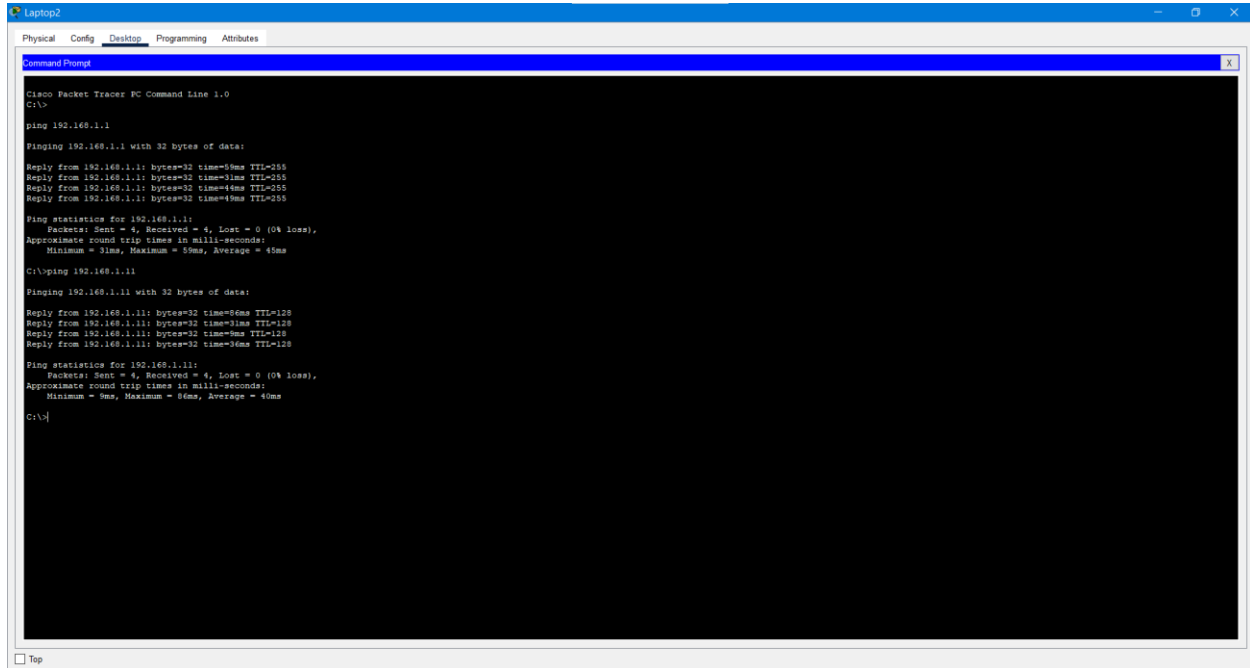


Reconnect the WIFI and get IPs via DHCP:



Verifying Connectivity:

- ping 192.168.1.1 → Successfully reached the default gateway (Cisco router)
- ping 192.168.1.11 → Communicated with PC2



```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=5ms TTL=255
Reply from 192.168.1.1: bytes=32 time=3ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 5ms, Average = 4ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128
Reply from 192.168.1.11: bytes=32 time=3ms TTL=128
Reply from 192.168.1.11: bytes=32 time=5ms TTL=128
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 4ms
C:\>
```

Wireless connectivity was stable within a reasonable range of the access point. Devices near the WRT300N connected instantly and received DHCP addresses. No dropouts were observed during testing.

Conclusion & Summary

The aim of this project was to design, configure, and test a basic wired and wireless network for a small office using Cisco Packet Tracer. The network was expected to support file sharing, internet access (simulated), mobile connectivity, and internal communication among devices.

What Was Done:

- Designed a star topology combining both wired and wireless devices.
- Used a Cisco router (192.168.1.1) as the default gateway and DHCP server.
- Connected multiple PCs via a switch and configured them to obtain IP addresses via DHCP.

- Installed and configured a WRT300N wireless router as an access point (192.168.1.2) to extend wireless coverage for laptops and smartphones.
- Added and configured a server (192.168.1.5) to simulate internal services like web hosting.
- Verified full connectivity using ping and HTTP tests.
- Ensured all devices received IP addresses from the main router's DHCP pool.

Why These Devices and Configurations Were Used:

- **Cisco Router:** Acts as a gateway, DHCP server, and central control point.
- **Switch:** Centralized connection point for all wired devices.
- **Wireless Router (WRT300N):** Used in access point mode to enable Wi-Fi without routing conflicts.
- **DHCP:** Simplifies IP address assignment and centralizes management.
- **Server:** Simulates internal services like HTTP access.

Outcome:

All devices — including PCs, laptops, smartphones, and the server — successfully connected to the network and communicated with each other. Wireless devices obtained correct IPs from the router. The web server responded instantly from client machines, demonstrating complete functionality of the designed network.